


```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh Ch01InChap01.dd
-rw-rw-r-- 1 kali kali 1.5M Jul  8 10:20 Ch01InChap01.dd

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ file Ch01InChap01.dd
Ch01InChap01.dd: DOS/MBR boot sector, code offset 0x34+2, OEM-ID ", 'k0nIHC" cached by Windows 9M, root entries 224, sectors 2880 (volumes ≤32 MB), sectors/FAT 9, sectors/track 18, dos < 4.0 BootSector (0), FAT (12 bit by descriptor+sectors), followed by FAT

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum Ch01InChap01.dd | tee ../reports/image_md5.txt
a117773bcf1fc88ec0ab8e0a349fbccb  Ch01InChap01.dd

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sha256sum Ch01InChap01.dd | tee ../reports/image_sha256.txt
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122  Ch01InChap01.dd

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/image_md5.txt
a117773bcf1fc88ec0ab8e0a349fbccb  Ch01InChap01.dd

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/image_sha256.txt
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122  Ch01InChap01.dd

Activate Windows
Go to Settings to activate Windows.
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ~/CIP-B102-Lab4/reports
total 8.0K
-rw-rw-r-- 1 kali kali 50 Jul  8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali 82 Jul  8 10:21 image_sha256.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

```
http://localhost:9999/autopsy
```

Keep this process running and use <ctrl-c> to exit

Can't open log: autopsy.log at /usr/share/autopsy/lib/Print.pm line 383.

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ sudo autopsy
```

```
[sudo] password for kali:
```

Autopsy Forensic Browser
http://www.sleuthkit.org/autopsy/
ver 2.24

Evidence Locker: /var/lib/autopsy

Start Time: Wed Jul 8 10:36:46 2026

Remote Host: localhost

Local Port: 9999

Open an HTML browser on the remote host and paste this URL in it:

```
http://localhost:9999/autopsy
```

Keep this process running and use <ctrl-c> to exit

← → ↻ 🏠 http://localhost:9999/autopsy?mod=0&view=1 ☆

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

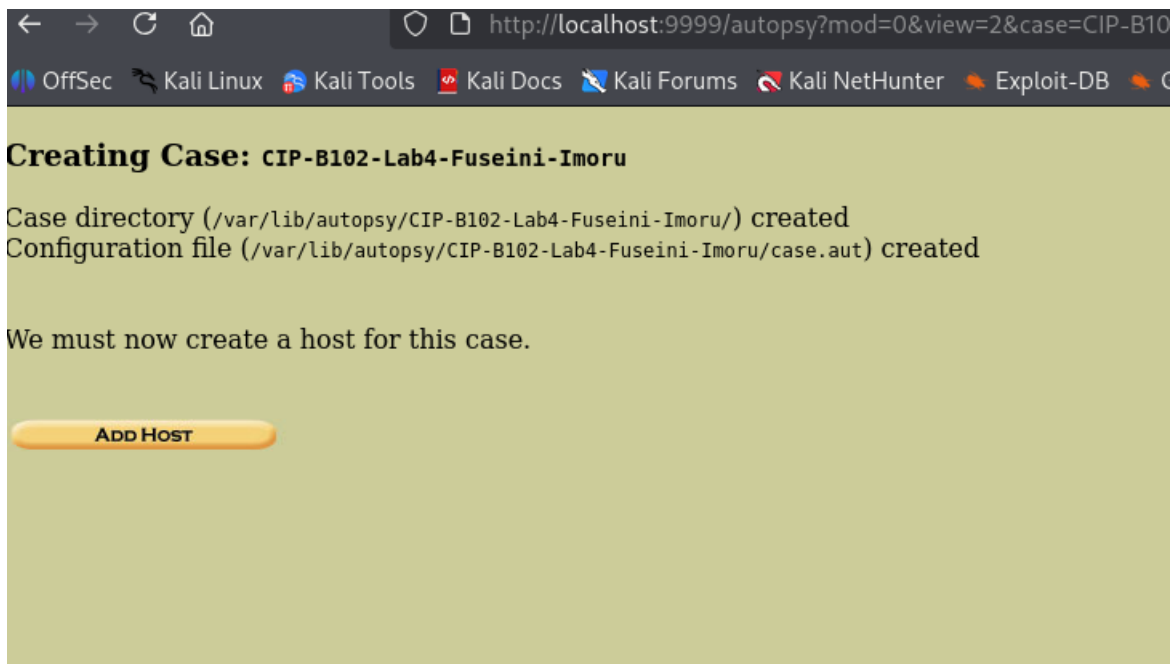
CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.

2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

a. Fuseini Imoru	b.
c.	d.
e.	f.
g.	h.
i.	j.



1. **Host Name:** The name of the computer being investigated. It can contain only letters, numbers, and symbols.

2. **Description:** An optional one-line description or note about this computer.

3. **Time zone:** An optional timezone value (i.e. EST5EDT). If not given, it defaults to the local setting. A list of time zones can be found in the help files.

4. **Timeskew Adjustment:** An optional value to describe how many seconds this computer's clock was out of sync. For example, if the computer was 10 seconds fast, then enter -10 to compensate.

5. **Path of Alert Hash Database:** An optional hash database of known bad files.

6. **Path of Ignore Hash Database:** An optional hash database of known good files.

Adding host: host1 to case CIP-B102-Lab4-Fuseini-Imoru

Host Directory (/var/lib/autopsy/CIP-B102-Lab4-Fuseini-Imoru/host1/) created

Configuration file (/var/lib/autopsy/CIP-B102-Lab4-Fuseini-Imoru/host1/host.aut) created

We must now import an image file for this host

ADD IMAGE

1. Location

Enter the full path (starting with /) to the image file.

If the image is split (either raw or EnCase), then enter '*' for the extension.

/me/kali/CIP-B102-Lab4/evidence/Ch01InChap01.dd

2. Type

Please select if this image file is for a disk or a single partition.

☒ Disk ☐ Partition

3. Import Method

To analyze the image file, it must be located in the evidence locker. It can be imported from its current location using a symbolic link, by copying it, or by moving it. Note that if a system failure occurs during the move, then the image could become corrupt.

☒ Symlink ☐ Copy ☐ Move

NEXT

CANCEL

HELP

Image File Details

Local Name: images/Ch01InChap01.dd

Data Integrity: An MD5 hash can be used to verify the integrity of the image. (With split images, this hash is for the full image file)

- ☐ Ignore the hash value for this image.
- ☒ Calculate the hash value for this image.
- ☐ Add the following MD5 hash value for this image:

☒ Verify hash after importing?

File System Details

Analysis of the image file shows the following partitions:

Partition 1 (Type: fat12)

Mount Point:

File System Type:

ADD

CANCEL

HELP

Calculating MD5 (this could take a while)
Current MD5: A117773BCF1FC88EC0A8E0A349FBBCB
Testing partitions
Linking image(s) into evidence locker
Image file added with ID img1

Volume image (0 to 0 - fat12 - C:) added with ID vol1

OK

ADD IMAGE

FILE ANALYSISKEYWORD SEARCHFILE TYPEIMAGE DETAILSMETA DATADATA UNITHELP?CLOSEX

Directory Seek

Enter the name of a directory that you want to view.
C:/

VIEW

File Name Search

Enter a Perl regular expression for the file names you want to find.

SEARCH

ALL DELETED FILES

EXPAND DIRECTORIES

All Deleted Files

Type	NAME	WRITTEN	ACCESSED	CREATED	SIZE	UID	GID	META
r / r	C:/Billing_letter.doc	2005-12-09 06:50:28 (EST)	2005-12-09 00:00:00 (EST)	2005-12-09 06:59:05 (EST)	24064	0	0	8
r / r	C:/confirmation.txt	2005-12-09 06:52:58 (EST)	2005-12-09 00:00:00 (EST)	2005-12-09 06:59:06 (EST)	227	0	0	11
r / r	C:/letter1.txt	2005-12-09 06:51:50 (EST)	2005-12-09 00:00:00 (EST)	2005-12-09 06:59:09 (EST)	121	0	0	15
r / r	C:/Regrets.doc	2005-12-09 06:50:52 (EST)	2005-12-09 00:00:00 (EST)	2005-12-09 06:59:09 (EST)	23552	0	0	17

File Browsing Mode

In this mode, you can view file and directory contents.
File contents will be shown in this window.
More file details can be found using the Metadata link at the end of the list (on the right).
You can also sort the files using the column headers

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ~/CIP-B102-Lab4/recovered
total 28K
-rw-rw-r-- 1 kali kali 24K Jul  8 13:10 vol1-C..Billing.Letter.doc
-rw-rw-r-- 1 kali kali 227 Jul  8 13:11 vol1-C..confirmation.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

Searching for ASCII: Done

Saving: Done

1 hits- [link to results](#)

Searching for Unicode: Done

Saving: Done

0 hits

[New Search](#)

1 occurrence of letter was found

Search Options:

ASCII

Case Sensitive

Sector 241 ([Hex](#) - [Ascii](#))

1: 59 (this letter, the)

letter was not found

Search Options:

Unicode

Case Sensitive

Searching for ASCII: Done

Saving: Done

1 hits- [link to results](#)

Searching for Unicode: Done

Saving: Done

0 hits

[New Search](#)

1 occurrence of password was found

Search Options:

ASCII

Case Sensitive

Sector 284 ([Hex](#) - [Ascii](#))

1: 167 (password: 342)

password was not found

Search Options:

Unicode

Case Sensitive

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ cd ~/CIP-B102-Lab4/evidence
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ img_stat Ch01InChap01.dd | tee ../reports/01_img_stat.txt
```

IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560

Sector size: 512

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ cat ../reports/01_img_stat.txt
```

IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560

Sector size: 512


```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/01_img_stat.txt
IMAGE FILE INFORMATION
```

Image Type: raw

Size in bytes: 1474560

Sector size: 512

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ mmls Ch01InChap01.dd | tee ../reports/02_mmls.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ fls Ch01InChap01.dd | tee ../reports/04_fls_root.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/04_fls_root.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ fls -r Ch01InChap01.dd | tee ../reports/05_fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ cat ../reports/05_fls_recursive.txt
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
```

```
└─$ fls -r -d Ch01InChap01.dd | tee ../reports/06_deleted_files.txt
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ cat ../reports/06_deleted_files.txt
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
└─$ ls -lh ../reports
total 24K
-rw-rw-r-- 1 kali kali 125 Jul  8 13:27 01_img_stat.txt
-rw-rw-r-- 1 kali kali  0 Jul  8 13:27 02_mmls.txt
-rw-rw-r-- 1 kali kali 215 Jul  8 13:30 04_fls_root.txt
-rw-rw-r-- 1 kali kali 215 Jul  8 13:30 05_fls_recursive.txt
-rw-rw-r-- 1 kali kali  99 Jul  8 13:31 06_deleted_files.txt
-rw-rw-r-- 1 kali kali  50 Jul  8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali  82 Jul  8 10:21 image_sha256.txt
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ fls -r Ch01InChap01.dd | grep -i "INCOME"
r/r 13: Income.xls
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ icat Ch01InChap01.dd 13 > ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ../recovered/INCOME_icat.xls
-rw-rw-r-- 1 kali kali 14K Jul  8 13:39 ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_md5.txt
6a2e65afc5af4fc5f9da2859df134eac  ../recovered/INCOME_icat.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sha256sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_sha256.txt
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d  ../recovered/INCOME_icat.xls
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ file ../recovered/INCOME_icat.xls
../recovered/INCOME_icat.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec  9 14:52:18 2005, Security: 0

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ istat Ch01InChap01.dd 13 | tee ../reports/income_istat.txt
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ cat ../reports/income_istat.txt
```

Directory Entry: 13

Allocated

File Attributes: File, Archive

Size: 13824

Name: INCOME.XLS

Directory Entry Times:

Written: 2005-12-09 06:52:18 (EST)

Accessed: 2005-12-09 00:00:00 (EST)

Created: 2005-12-09 06:59:07 (EST)

Sectors:

285 286 287 288 289 290 291 292

293 294 295 296 297 298 299 300

301 302 303 304 305 306 307 308

309 310 311

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ : > ../recovered/INCOME_blkcat.xls
```

```
for blk in 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
307 308 309 310 311; do
  blkcat Ch01InChap01.dd "$blk" >> ../recovered/INCOME_blkcat.xls
done
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ ls -lh ../recovered/INCOME_blkcat.xls
```

```
-rw-rw-r-- 1 kali kali 14K Jul  8 13:47 ../recovered/INCOME_blkcat.xls
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ file ../recovered/INCOME_blkcat.xls
```

```
../recovered/INCOME_blkcat.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Ve
rsion 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creat
ing Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date:
Fri Dec  9 14:52:18 2005, Security: 0
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ md5sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_md5.txt
```

```
6a2e65afc5af4fc5f9da2859df134eac ../recovered/INCOME_blkcat.xls
```

Activate Windows

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ md5sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_md5.txt
```

```
6a2e65afc5af4fc5f9da2859df134eac ../recovered/INCOME_blkcat.xls
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

```
$ sha256sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_sha256.txt
```

```
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d ../recovered/INCOME_blkcat.xls
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```

Activate Windows


```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ mkdir -p ../recovered/tsk_recover_all

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ tsk_recover -e Ch01InChap01.dd ../recovered/tsk_recover_all
Files Recovered: 6

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ find ../recovered/tsk_recover_all -iname "*INCOME*" -ls | tee ../reports/income_tsk_recover_location.txt
  525188      16 -rw-rw-r--  1 kali   kali      13824 Jul  8 13:58 ../recovered/tsk_recover_all/Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/tsk_recover_all/*INCOME* 2>/dev/null | tee ../reports/income_tsk_recover_md5.txt

```

```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ md5sum ../recovered/INCOME_icat.xls
md5sum ../recovered/INCOME_blkcat.xls
find ../recovered/tsk_recover_all -iname "*INCOME*" -exec md5sum {} \;
6a2e65afc5af4fc5f9da2859df134eac ../recovered/INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac ../recovered/INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac ../recovered/tsk_recover_all/Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ libreoffice ../recovered/INCOME_icat.xls
Command 'libreoffice' not found, but can be installed with:
sudo apt install libreoffice-common
Do you want to install it? (N/y)y
sudo apt install libreoffice-common
[sudo] password for kali:
The following package was automatically installed and is no longer required:
  libvidstab1.1
Use 'sudo apt autoremove' to remove it.

Installing:
  libreoffice-common

Installing dependencies:

```

```

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ strings -a Ch01InChap01.dd | grep -i "amelia" | tee ../reports/search_amelia.txt
Amelia Phillips
Amelia PhillipsKKD
Amelia PhillipsKKD
Amelia Phillips
Amelia Phillips
Amelia Phillips
Amelia Phillips
  B
Amelia Phillips
Amelia Phillips
Amelia Phillips
Amelia Phillips

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sudo mkdir -p /mnt/cipb102_lab4
[sudo] password for kali:

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sudo mount -o loop,ro ~/CIP-B102-Lab4/evidence/Ch01InChap01.dd /mnt/cipb102_lab4

```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -la /mnt/cipb102_lab4 | tee ~/CIP-B102-Lab4/reports/mounted_listing.txt
total 127
drwxr-xr-x 2 root root 7168 Dec 31 1969 .
drwxr-xr-x 4 root root 4096 Jul 8 15:21 ..
-rwxr-xr-x 1 root root 104448 Dec 9 2005 Client Info.mdb
-rwxr-xr-x 1 root root 13824 Dec 9 2005 Income.xls

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ sudo umount /mnt/cipb102_lab4
```

```
(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
$ ls -lh ~/CIP-B102-Lab4/reports
total 56K
-rw-rw-r-- 1 kali kali 125 Jul 8 13:27 01_img_stat.txt
-rw-rw-r-- 1 kali kali 0 Jul 8 13:27 02_mmls.txt
-rw-rw-r-- 1 kali kali 215 Jul 8 13:30 04_fls_root.txt
-rw-rw-r-- 1 kali kali 215 Jul 8 13:30 05_fls_recursive.txt
-rw-rw-r-- 1 kali kali 99 Jul 8 13:31 06_deleted_files.txt
-rw-rw-r-- 1 kali kali 50 Jul 8 10:21 image_md5.txt
-rw-rw-r-- 1 kali kali 82 Jul 8 10:21 image_sha256.txt
-rw-rw-r-- 1 kali kali 65 Jul 8 13:50 income_blkcat_md5.txt
-rw-rw-r-- 1 kali kali 97 Jul 8 13:50 income_blkcat_sha256.txt
-rw-rw-r-- 1 kali kali 63 Jul 8 13:40 income_icat_md5.txt
-rw-rw-r-- 1 kali kali 95 Jul 8 13:40 income_icat_sha256.txt
-rw-rw-r-- 1 kali kali 342 Jul 8 13:44 income_istat.txt
-rw-rw-r-- 1 kali kali 112 Jul 8 13:58 income_tsk_recover_location.txt
-rw-rw-r-- 1 kali kali 0 Jul 8 13:58 income_tsk_recover_md5.txt
-rw-rw-r-- 1 kali kali 214 Jul 8 15:22 mounted_listing.txt
-rw-rw-r-- 1 kali kali 261 Jul 8 15:21 search_amelia.txt

(kali㉿kali)-[~/CIP-B102-Lab4/evidence]
```